Unit: Unit 0: Foundations of Algebra
Topic: GCF LCM

Name: ________________
Date: ____________

Advanced (Carolina Reaper)

Q1. Find the GCF of 12 and 18 using factors.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q2. Find the LCM of 9 and 12 using multiples.
(A) 18
(B) 27
(C) 36
(D) 45
(E) 54

Q3. Find the GCF of 24 and 30 using prime factorization.
(A) 2
(B) 3
(C) 6
(D) 12
(E) 24

Q4. A bookshelf has 12 shelves, and each shelf can hold 8 books. If the books are packed in boxes that hold 24 books, how many boxes are needed to fill the bookshelf?
(A) 4
(B) 6
(C) 8
(D) 12
(E) 16

Q5. Find the LCM of 8 and 15 using prime factorization.
(A) 24
(B) 30
(C) 40
(D) 60
(E) 120

Q6. A bakery produces 18 cupcakes per hour, and a store sells 12 cupcakes per hour. If they want to package the cupcakes in boxes that hold the same number of cupcakes, what is the smallest number of cupcakes per box?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 18

Q7. Find the GCF of 20 and 25 using factors.
(A) 1
(B) 5
(C) 10
(D) 15
(E) 20

Q8. A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces of candy will each friend get?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 12

Open Ended Questions (FRQ)

Q9. Find the GCF and LCM of 18 and 24 using prime factorization. Show your work and explain your reasoning.

Q10. A water tank can hold 120 gallons of water. If 3 pipes can fill the tank at a rate of 8 gallons per minute, and 2 pipes can empty the tank at a rate of 4 gallons per minute, how many minutes will it take to fill the tank? Use GCF and LCM to help you solve the problem.

Chef's Tips

- Remind students to use prime factorization to find GCF and LCM.
- Encourage students to show their work and explain their reasoning.
- Emphasize the importance of using correct units and labels in their answers.